Zepeng Zhang

Intelligent Maintenance and Operations Systems (IMOS) Lab
École Polytechnique Fédérale de Lausanne (EPFL)

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EDUCATION

Ph.D. Electrical Engineering, École Polytechnique Fédérale de Lausanne, 2023.09-present
Supervisor: Prof. Olga Fink

M.S. Information and Communication Engineering, ShanghaiTech University, 2020.09-2023.07
Thesis: Optimization-Induced Graph Neural Networks

B.Eng. Measuring & Control Technology and Instrumentations, Wuhan University, 2016.09-2020.05
Thesis: A Deep Learning-Aided Approach to Portfolio Design for Financial Index Tracking

EXPERIENCES

2025.04–2025.08 Visiting Student, Télécom Paris, Institut Polytechnique de Paris

2023.02–2023.04 Research Intern, Peking University

2019.12–2020.09 Visiting Student, ShanghaiTech University

2019.07–2019.09 Visiting Student, City Univercity of HongKong

2018.01–2020.06 Member, WHUAI (ZhuYing Team)

CURRENT RESEARCH INTERESTS

Graph Machine Learning, Generative Models, Graph Signal Processing, Domain Adaptation

PUBLICATIONS

1. Graph-Spectral Flow Matching for Multivariate Time Series Anomaly Detection
Zepeng Zhang, Jhony H. Giraldo, Wenbin Wang and Olga Fink
Under Review

2. GSLAD: Prototype-Regularized Graph Structure Learning for Multivariate Time Series Anomaly Detection
Zepeng Zhang, Fuad Khuri, Keivan Faghih Niresi and Olga Fink
Under Review

3. Spatiotemporal Imputation with Graph-Informed Flow Matching
Zepeng Zhang, Aref Einizade, Jhony H. Giraldo and Olga Fink
International Conference on Machine Learning (ICML) 2026

4. From Physics to Machine Learning and Back: Part I - Learning with Inductive Bias in Prognostics and Health Management (PHM)

Olga Fink, Vinay Sharma, Ismail Nejjar, Leandro Von Krannichfeldt, Sergei Garmaev, **Zepeng Zhang**, Amaury Wei, Gaetan Frusque, Florent Forest, Mengjie Zhao, Chi-Ching Hsu, Keivan Faghih Niresi, Han Sun, Hao Dong, Chenghao Xu, Raffael Theiler, Arthur Bizzi, Kevin Steiner
Reliability Engineering & System Safety (RESS) 2026

5. From Physics to Machine Learning and Back: Part II - Learning and Observational Bias in Prognostics and Health Management (PHM)
Olga Fink, Ismail Nejjar, Vinay Sharma,Keivan Faghih Niresi, Han Sun, Hao Dong, Chenghao Xu, Amaury Wei, Arthur Bizzi, Raffael Theiler, Yuan Tian, Leandro Von Krannichfeldt, Zhan Ma, Sergei Garmaev, **Zepeng Zhang**, Mengjie Zhao, Kevin Steiner, Yusuf Kesmen
Reliability Engineering & System Safety (RESS) 2026

6. MM-Net: Recurrent Unfolding with Adaptive Majorization for Weighted Sum-Rate Beamforming
Zhexian Yang, **Zepeng Zhang** and Ziping Zhao
IEEE Signal Processing Letters 2026

7. Weighted Sum Secrecy Rate Maximization via Minorization-Maximization
Zhexian Yang, **Zepeng Zhang** and Ziping Zhao
IEEE International Symposium on Information Theory (ISIT) 2026

8. Domain Adaptive Unfolded Graph Neural Networks
Zepeng Zhang and Olga Fink
AAAI Conference on Artificial Intelligence (AAAI) 2025

9. Algorithm-Informed Graph Neural Networks for Leakage Detection and Localization in Water Distribution Networks
Zepeng Zhang and Olga Fink
Reliability Engineering & System Safety (RESS) 2025

10. RINS-T: Robust Implicit Neural Solvers for Time Series Linear Inverse Problems
Keivan Faghih Niresi, **Zepeng Zhang** and Olga Fink
IEEE Transactions on Instrumentation and Measurement (TIM) 2025

11. GeoLink: Empowering Remote Sensing Foundation Model with OpenStreetMap Data
Lubin Bai, Xiuyuan Zhang, Siqi Zhang, **Zepeng Zhang**, Haoyu Wang, Wei Qin, and Shihong Du
Conference on Neural Information Processing Systems (NeurIPS) 2025

12. Weighted Sum-Rate Maximization for Beamforming Design Using Minorization-Maximization: Convergence Rate and Deep Unfolding
Zhexian Yang, **Zepeng Zhang**, and Ziping Zhao
European Signal Processing Conference (EUSIPCO) 2025

13. Graph Neural Networks With Adaptive Structures
Zepeng Zhang, Songtao Lu, Zengfeng Huang, and Ziping Zhao
IEEE Journal of Selected Topics in Signal Processing: Special Series on AI in Signal & Data Science (JSTSP) 2024

14. Accelerating Quadratic Transform and WMMSE
Kaiming Shen, Ziping Zhao, Yannan Chen, **Zepeng Zhang**, and Hei Victor Cheng
IEEE Journal on Selected Areas in Communications: Special Issue on Advanced Optimization Theory and Algorithms for Next Generation Wireless Communication Networks (JSAC) 2024

15. A Joint Graph Signal and Laplacian Denoising Network
Zepeng Zhang and Ziping Zhao
Asia-Pacific Signal and Information Processing Conference (ACSSC/Asilomar) 2024

16. Accelerating Quadratic Transform and WMMSE
Kaiming Shen, Ziping Zhao, Yannan Chen, **Zepeng Zhang**, and Hei Victor Cheng
IEEE International Symposium on Information Theory (ISIT) 2024

17. Enhancing the Efficiency of WMMSE and FP for Beamforming by Minorization-Maximization
Zepeng Zhang, Ziping Zhao, and Kaiming Shen
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2023

18. Towards Understanding Graph Neural Networks: An Algorithm Unrolling Perspective
Zepeng Zhang and Ziping Zhao
International KDD Workshop on Deep Learning on Graphs: Methods and Applications 2022

19. Weighted Sum-Rate Maximization for Multi-Hop RIS-Aided Multi-User Communications: A Minorization-Maximization Approach
Zepeng Zhang and Ziping Zhao
IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC) 2021
Best Student Paper Award Finalist

20. Vast Portfolio Selection With Submodular Norm Regularizations
Zepeng Zhang, and Ziping Zhao
European Signal Processing Conference (EUSIPCO) 2021

21. Multi-Period Portfolio Optimization for Financial Index Tracking
Xiuyuan Huang, **Zepeng Zhang**, and Ziping Zhao
Asilomar Conference on Signals, Systems, and Computers (ACSSC/Asilomar) 2021

22. Scalable Financial Index Tracking With Graph Neural Networks
Zepeng Zhang, and Ziping Zhao
IEEE Statistical Signal Processing Workshop (SSP) 2021

23. A Deep Learning-Aided Approach to Portfolio Design for Financial Index Tracking
Zepeng Zhang, and Ziping Zhao
Asilomar Conference on Signals, Systems, and Computers (ACSSC/Asilomar) 2020

AWARDS AND HONORS

2021 Best Student Paper Award Finalist in IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)

2019 Yu Gang Scholarship (Top 1%, Rank 1st out of the department), WHU
Outstanding Student First-Class Scholarship (Top 5%), WHU

2018 SAMSUNG Scholarship (Top 1%, Rank 1st out of the department), WHU
Outstanding Student First-Class Scholarship (Top 5%), WHU
1st prize of RoboCup China Open
1st prize of Asia and Pacific Mathematical Contest in Modeling
3rd prize of The Chinese Robot Skills Competitions

2017 Outstanding Student Second-Class Scholarship (Top 15%), WHU
1st prize of The Chinese Mathematics Competitions, Hubei Province

TEACHING ASSISTANT

EPFL

CIVIL 426: Machine learning for predictive maintenance applications 2023-24 Fall, 2024-25 Fall, 2025-26 Fall, 2026-27 Fall

CIVIL 332: Data Science for infrastructure condition monitoring 2023-24 Spring, 2024-25 Spring, 2025-26 Spring

MATH 106: Math Analyse II, 2024-25 Spring

ShanghaiTech

CS182: Introduction to Machine Learning 2021-22 Fall

SI 231: Matrix Analysis and Computations 2020-21 Spring, 2021-22 Spring

ACADEMIC SERVICE

Reviewer

Journal Reviewer: IEEE Transactions on Wireless Communications, Mechanical Systems and Signal Processing, IEEE Open Journal of Signal Processing, Neurocomputing

Conference Reviewer: ICML, ICLR, NeurIPS, LOG, AISTATS, WWW, ICASSP, ECAI, MLSP

Student Supervision

Fuad Khuri: Regime-Conditioned Graph-Informed Flow Matching for Multivariate Time Series Anomaly Detection

Fuad Khuri: Graph Structure Learning for Fault Detection in Multivariate Time Series

Moein Nasiri: Rethinking Unfolded Graph Neural Networks

Yuhao Liu: Graph Neural Networks for Cascading Failure Analysis in Power Grids

Xiao Tan: Aligning Graph Neural Networks and Large Language Models for Learning on Text-Attributed Graphs

Conference Organizer

Intelligent Maintenance Conference (IMC) 2023, 2024, 2025, 2026

SKILLS

Python, MATLAB, CVX, PyTorch, PyG, LaTeX

Toefl: 105 (Reading: 30; Listening: 27; Speaking: 22; Writing: 26)